CHAPTER ONE: INTRODUCTION TO STATISTICS

1. DEFINITION & MEANING OF STATISTICS

Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data. In other words, it is a mathematical discipline to collect and summarize data. Also, we can say that statistics is a branch of applied mathematics. We can also say that statistics is a branch of mathematics that deals with the collection, analysis, interpretation, presentation, and organization of data. It is a tool used to understand and analyze data in various fields such as economics, business, engineering, healthcare, social sciences, and more. The primary objective of statistics is to convert raw data into meaningful information, helping individuals and organizations make informed decisions. According to statistician Arthur Lyon Bowley, Statistics is defined as "Numerical statements of facts in any department of inquiry placed in relation to each other.

For example, suppose you need to find how many people are employed in Chattogram city. Since the city is populated with 2 crore, hence we will take a survey here for 1000 people (sample). Based on that, we will create the data, which is the statistics.

Meaning of Statistics in Sharia Research

Statistics in Sharia research can be used to analyze legal rulings, practices, and societal impacts of Sharia law. It helps researchers make sense of qualitative and quantitative data, enabling them to assess the extent of implementation and societal effects, as well as to draw conclusions about its influence on different communities.

For example, a study on the impact of Islamic finance (a practice governed by Sharia law) on economic growth in Islamic countries can use statistics to analyze data such as the number of Islamic financial institutions, market share, and financial growth rates, thereby identifying patterns and trends.

2. TYPES OF STATISTICS

Basically, there are two types of statistics.

2.1 Descriptive Statistics
2.2 Inferential Statistics

2.1 Descriptive Statistics

The data is summarized and explained in descriptive statistics. The summarization is done from a population sample utilizing several factors such as mean and standard deviation. Descriptive statistics is a way of organizing, representing, and explaining a set of data using charts, graphs, and summary measures. Histograms, pie charts, bars, and scatter plots are common ways to summarize data and present it in tables or graphs. This includes measures like:

- **Central Tendency** (mean, median, mode)
- **Dispersion** (range, variance, standard deviation)
- **Graphs** (bar graphs, histograms, pie charts, etc.)

For example, if we have the data on the number of hours spent by 100 students on studying each week, we can use descriptive statistics to calculate the average study time (mean), the most common study time (mode), and the range of study times.

In Sharia research, descriptive statistics can be used to summarize legal rulings, social practices, or survey data related to Islamic law. This can include the frequency of certain practices (e.g., zakat payments, fasting during Ramadan) and their impact on the community. *Example: Analyzing the number of people paying zakat (almsgiving) in a community over several years using statistical tools such as averages, percentages, and distributions.*

2.2 Inferential Statistics

We attempt to interpret the meaning of descriptive statistics using inferential statistics. We utilize inferential statistics to convey the meaning of the collected data after it has been collected, evaluated, and summarized. The probability principle is used in inferential statistics to determine if patterns found in a study sample may be extrapolated to the wider population from which the sample was drawn. Inferential statistics are used to test hypotheses and study correlations between variables, and they can also be used to predict population sizes. Inferential statistics are used to derive conclusions and inferences from samples, i.e. to create accurate generalization. In Shari'ah research, Inferential statistics would allow researchers to make generalizations about the implementation of Sharia laws in different regions or populations. For instance, researchers can use samples from different Islamic countries to infer the broader impact of Sharia governance.

For example, estimating the effect of Sharia law on the reduction of crime rates in countries where Sharia-based legal systems are applied, by using a sample of data from a few countries to make inferences about the global situation.

3. Functions of Statistics

a. Presents facts in a definite form.
b. Simplifies mass of figures.
c. It facilitates comparison.
d. It helps in formulating and testing hypothesis.
e. It helps in prediction.
f. It helps in the formulation of suitable policies.

a) Definiteness: Numerical expressions are convincing and, therefore, one of the most important functions of statistics is to present general statements in a precise and definite form.So, Statistics presents facts in clear, numerical terms, removing vagueness. For example,instead of saying *"Many people in Bangladesh live in poverty"*, statistics makes it precise: *"According to BBS (2022), 18.7% of the population lives below the poverty line."*. Instead of saying "Most Muslims pay Zakat regularly", a statistical survey can show "67% of eligible Muslims in Dhaka reported paying Zakat in 2024." This gives definite evidence to scholars and policymakers.

b) Simplifies mass of figures: Not only does Statistics present facts in a definite form but it also helps in simplify mass of data into a few significant figures. Large data is condensed into simple figures, averages, or ratios. Recording the daily income of every rickshaw puller in Dhaka would be impossible. But statistics condense it: *"The average daily income of a rickshaw puller is Tk. 650."* Instead of collecting every family's charity spending in Ramadan, statistics can condense it: *"On average, a Muslim household in Chattogram spends Tk. 4,500 for Sadaqah and Zakat during Ramadan."*

c) Comparison: Statistics allows meaningful comparisons across time, region, or group. For example, It is not enough to say "Education has improved." A meaningful comparison is: "Literacy rate increased from 58% in 2010 to 76% in 2023." Comparing the practice of Islamic banking with conventional banking: "Islamic banks in Bangladesh captured 25% of deposits in 2010, which increased to nearly 36% in 2022." This comparison shows the growing preference for Shariah-compliant finance.

d) **Formulating and hypothesis testing:** Hypothesis. Statistical methods are extremely useful in formulating and testing hypothesis and to develop new theories. *"Digital payment adoption reduces corruption."* Researchers test this by analyzing transaction data across mobile banking platforms. *Islamic microfinance reduces rural poverty more effectively than conventional microfinance."* By comparing income levels of beneficiaries under both systems, researchers can statistically verify this.

e) **Prediction:** Statistical methods provide helpful means of forecasting future events.Past data helps forecast future outcomes. For example, if a businessman has to decide how much he should produce in 2026, he would like to know the expected sales for that year. However, a better method for him would be to analyse the sales data of the past years or arrange a statistical survey of the market to obtain necessary data for estimating the sales volume for the year 2026. By analyzing past Zakat collections in Bangladesh, predictions can be made for future years. For instance: *“Based on past trends, Zakat collection in Bangladesh is expected to cross Tk. 500 billion by 2030.”* Such forecasts can guide Shariah-based poverty reduction programs.

f) Formulation of policies. Statistics provide the basic material for framing suitable policies. The government needs statistics on unemployment to frame job creation policies. For instance: *“Youth unemployment in Bangladesh was 10.6% in 2022.”* Based on this, skill development programs are designed. If statistics show that *“only 40% of Muslim-owned businesses follow Islamic finance contracts (Murabaha, Musharakah, etc.),”* then policymakers can design awareness campaigns and Shariah-compliant financial products to increase compliance.

Summary Table: Bangladesh & Shariah Research Examples

Function	Bangladesh Example	Shariah Research Example
Definiteness	Poverty rate = 18.7%	67% Muslims in Dhaka paid Zakat
Condensation	Avg. rickshaw puller earns Tk. 650/day	Avg. family donates Tk. 4,500 in Ramadan
Comparison	Literacy rate: 58% (2010) → 76% (2023)	Islamic banking share: 25% (2010) → 36% (2022)
Hypothesis Testing	Mobile payments reduce corruption	Islamic microfinance reduces poverty
Prediction	Forecast rice production for 2030	Zakat collection may cross Tk. 500B by 2030
Policy Formulation	Youth unemployment = 10.6% → skill programs	Promote Shariah-compliant banking via awareness

4. CHARACTERISTICS OF STATISTICS

❑ **Statistics deals with aggregate of individuals rather than with individual alone:** Single observation cannot be called as Statistics For example: Production of a food grain in a particular year, percentage marks of a single student can not be Statistics

❑ **Statistics varied by multiplicity of causes:** If we consider production of a rice in a Chattogram for few years, obviously the figures will differ in every year. This is because the Statistical data are affected to a marked extent by multiple causes. The causes for change in the production of rice may be rainfall, fertility of soil, seeds & fertilizers used, methods of cultivations etc.

❑ **Statistics should be expressed as numerical figures:** Statistical data should be expressed in terms of numbers. Production of sugar in Chattogram. Pass percentage of students at HSC Examination should be recorded over a period of time to constitute Statistical Data.

❑ **Statistics collected should be of reasonable standard of accuracy:** The accuracy standard should be pre decided.

❑ **Statistics are collected for a pre-determined purpose:** The purpose should be well defined and clear, otherwise some unnecessary information may be collected or necessary information may be ignored.

5. SAMPLE & POPULATION

- **Population:** A population is a set of units (usually people, objects, transactions, or events) that we are interested in studying**.** Example: The population could be all voters in a country. The population might be all Muslims in a particular country, and the study may aim to analyze how many of them follow Sharia-based financial systems.

- **Sample:** A sample is a subset of the units of a population. It is often impractical or impossible to collect data from the entire population, so a sample is used to make generalizations about the population. Example: A sample might consist of 1,000 voters chosen randomly from the larger population of millions of voters. A sample of 100 Sharia-compliant financial institutions could be analyzed to assess the growth of Islamic banking.

6. IMPORTANCE OF STATISTICS

Statistics is used to conduct research, evaluate outcomes, develop critical thinking, and make informed decisions about a set of data. Statistics can be used to inquire about almost any field of study to investigate why things happen, when they occur, and whether reoccurrence is predictable.

i) Statistics and the State

A state collects vast amounts of statistical data for governance, including information on prices, production, consumption, income, expenditure, investments, and profits. In Bangladesh, such as the Bangladesh Bureau of Statistics (BBS) play a central role in gathering and analyzing the data to formulate policies and implement various welfare programs. Widely used statistical methods, including time-series analysis, index numbers, and forecasting, are extensively employed to guide decisions in economic planning, resource allocation, and social development programs. These methods help the government in addressing national challenges and promoting sustainable growth in sectors such as agriculture, industry, and social welfare.

ii) **Statistics in Economics**

Statistical methods are integral to economics, used in:

- **Time-series analysis** to study price behavior, production, and consumption trends.
- **Index numbers** for tracking changes in prices, imports, exports, and industrial production.
- **Demand analysis** to understand the relationship between price and supply.
- **Forecasting** techniques for predicting inflation, unemployment, and capacity utilization.

iii) Statistics in Business Management

In business, statistics helps make informed decisions under uncertainty:

- **Marketing**: Statistics are used to analyze consumer behavior, purchasing power, and market trends to determine product viability.
- **Production**: Statistical methods are applied in R&D for quality control and to decide whether to manufacture in-house or outsource.
- **Finance**: Through **correlation analysis**, companies predict future dividends, assess financial health, and make investment decisions.
- **Personnel**: Statistical analysis aids in workforce planning, studying trends like wage rates, labor turnover, and training needs.

iv) Statistics in Physical Sciences

In physical sciences (e.g., **astronomy, engineering, meteorology**), statistical techniques like **sampling** and **experimental design** help in analyzing complex data, aiding in discoveries and advancements.

v) **Statistics in Social Sciences**

Statistics is essential in social sciences to measure and analyze societal phenomena:

- **Regression and correlation analysis** help study relationships between social variables like income, education, and crime.
- **Sampling and estimation** are used to conduct social surveys and make valid inferences about communities.
- In **sociology**, statistics is used to analyze vital statistics like **birth and death rates** and **population growth**.

vi) Statistics in Medical Sciences

In medicine, statistics is used to analyze health data and determine the efficacy of treatments:

- **Statistical tests** (e.g., **t-tests** and **F-tests**) are used to compare the effectiveness of drugs.
- Health data such as **pulse rate**, **blood pressure**, and **temperature** are used to monitor and diagnose diseases.

vii) Importance of Statistics in Shariah Research

In Shariah research, statistics can be vital for empirical studies and data-driven conclusions:

- Population studies: Statistical methods can be used to track the observance of Islamic practices (e.g., prayer, fasting) in different communities, revealing trends and societal behaviors.

  Example: A study on the frequency of Zakat (charity) payments across different regions can highlight socioeconomic trends and the effectiveness of charity programs in alleviating poverty. By using regression analysis, researchers can examine the impact of Zakat on improving community welfare, taking into account variables like income, education, and community development.

- Economic Studies: In analyzing Islamic finance, statistics can help assess the performance of Islamic banking systems, comparing profit-sharing models like Mudarabah with traditional interest-based systems.

  Example: A study could use statistical methods to compare the returns of Islamic investment portfolios to conventional ones, using forecasting techniques to predict future returns.

- Social Impact: Shariah-compliant policies like those related to inheritance or family law can be studied using statistical methods to assess their impact on society.

Example: A statistical survey could examine the impact of Shariah inheritance laws on wealth distribution and family dynamics, providing insights into how these laws affect economic equality across different regions.

7. LIMITATIONS OF STATISTICS

Despite the usefulness of statistics in many fields, impression should not be carried that statistics are like magical devices which always provide the correct solution of problems. Unless the data are properly collected and critically interpreted there is every likelihood of drawing wrong conclusions. Therefore, it is also necessary to know the limitations and the possible misuses of statistics. The following are the important limitations of the science of statistics:

i) **Statistics does not deal with isolated measurement.** Not all quantitative data are statistical. Isolated measurements are not statistical. Data are statistical when they relate to measurement of masses, not statistical when they relate to an individual item or event as a separate entity. For example, measuring the height of the Bangladesh Army and claiming it represents the height of the general population is not statistically valid. The correct approach would be to take a large sample of individuals from different age groups, genders, and ethnic backgrounds to obtain meaningful data. In Shariah studies, taking a single instance of a specific ruling or fatwa and generalizing it to apply to all similar situations without considering the context or other related factors can lead to wrong interpretations.

ii) **Statistics deals only with quantitative characteristics.** Statistics are numerical statements of facts. Such characteristics as cannot be expressed in numbers are incapable of statistical analysis. Thus, qualitative characteristics like honesty, efficiency, intelligence, blindness and deafness cannot be studied directly. For example, It cannot measure abstract qualities like the level of emotional connection a customer feels towards the brand like ILLIYEN. For example, measuring how many customers are satisfied with a service is a quantitative data point, but measuring customer loyalty or trust is qualitative and cannot be directly measured by statistics alone.In Shariah studies, a statistical approach may quantify the number of people who perform certain acts of worship (like prayer attendance) but cannot capture the quality of their spiritual connection or sincerity in performing these acts.

iii) **Statistical results are true only on an average.** The conclusions obtained statistically are not universally true they are true only under certain conditions. For example, A study showing that the average income of a country is 50,000 Taka does not mean every individual in that country earns 50,000 Taka. There could be people earning much less or much more. Similarly, the average test score of a class may be high, but some students may have failed, and others may have excelled, which would not be apparent from just the average score.In the context of Zakat (obligatory almsgiving in Islam), the average income might suggest that most people are eligible to give Zakat, but it does not highlight the specific cases where individuals might fall below the Nisab (minimum threshold for giving Zakat) despite the average being higher. Thus, relying on averages in this case may overlook people in need.

iv) **Statistics is only a means:** Relying solely on the statistical data might lead to misleading conclusions Statistical methods furnish only one method of studying a problem. They may not provide the best solution under all circumstances. It should be carefully noted that statistics is only a means and not an end. In deciding a course of action, it may be necessary to take into account the country's culture, religions, philosophy, personal, political or other non-quantitative considerations. Exclusive dependence on statistics may lead to fallacious conclusions in many situations. For example: If you were analyzing the effectiveness of a new medicine, statistics might show that 60% of patients report improvement. However, this number alone does not provide the full picture. To get a complete understanding, you would need to consider other factors, such as the patient's overall health, lifestyle, and underlying conditions, which statistics cannot fully capture. In Shariah studies, statistical methods might help in determining the number of people adhering to certain practices, such as fasting during Ramadan. For example, someone may physically fast, but if they are unable to perform the fast spiritually due to personal issues, this would not be captured in the statistical data.

v) **Statistics can be misused.** The greatest limitation of statistics is that it is liable to be misused. For example, if statistical conclusions are based on incomplete information, one may arrive at fallacious conclusions. For Example**,** Statistics can be easily manipulated to support a particular agenda. For example, a politician might claim that crime has decreased by 30% based on statistical data, but if the data only includes specific types of crimes (e.g., property crimes) and omits others (e.g., violent crimes), the conclusion is misleading. By selectively presenting data, the politician is misusing statistics to create a false narrative of safety, despite the reality being different. For example, statistics showing a decline in the number of women wearing hijabs might be used to argue against the requirement of hijab, ignoring the complexity of personal choice, culture, and individual circumstances. Without a deep understanding of Shariah principles, relying on statistics in this manner can lead to unjustified conclusions.

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